Extra Bite #8: Oscillators, Synthesizers, and PLL

Three classic oscillators, the PLL servo loop, direct digital synthesis, and how to keep a frequency from drifting.

Group E7H · 13 pool questions · Standalone · 5-10 minute presentation · Practice exams at eham.net/exams

The Three Classic Oscillators

Type	How the feedback gets back to the input	Q
Colpitts	Through a **capacitive divider** in the tank circuit.	E7H04
Hartley	Through a **tapped coil** (inductive divider). Not asked directly, but completes the trio.	—
Pierce	Through a **quartz crystal**. Very common in stable crystal oscillators.	E7H05

What are three common oscillator circuits? Colpitts, Hartley, and Pierce. (E7H01)

Phase-Locked Loop (PLL)

A PLL is a feedback loop that locks a voltage-controlled oscillator (VCO) to a stable reference frequency by adjusting the VCO until the phase comparison matches.

A PLL is an electronic servo loop consisting of a phase detector, a low-pass filter, a voltage-controlled oscillator, and a stable reference oscillator. (E7H03)

The PLL can do two jobs in a radio: (E7H06)

- **Frequency synthesis** – generate many precise output frequencies from one reference.
- **FM demodulation** – lock to an incoming FM signal; the VCO's control voltage IS the recovered audio.

Direct Digital Synthesis (DDS)

A DDS builds an output waveform by digitally stepping through a stored lookup table and feeding the values through a DAC. The classic block diagram has four parts:

Phase accumulator → lookup table → digital-to-analog converter → low-pass anti-alias filter. (E7H09)

Question	Answer	Q
What is in the DDS lookup table?	**Amplitude values that represent the desired waveform** (a sine wave's instantaneous values, for example).	E7H10
Major spectral impurity of a DDS?	**Spurious signals at discrete frequencies** – the spurs arise primarily from phase truncation (dropping LSBs of the accumulator before the lookup table) and from DAC quantization nonlinearity.	E7H11

Stability And Drift

Problem	Fix	Q
Thermal drift in a crystal oscillator	Use **NP0 capacitors** (C0G/NP0 class: nominally zero temperature coefficient, 0 ppm/°C).	E7H08
Microphonics (frequency wobble from vibration)	A microphonic is a **change in oscillator frequency caused by mechanical vibration**. (E7H02) Fix: **mechanically isolate the oscillator circuitry from its enclosure**. (E7H07)	E7H02, E7H07
Crystal not exactly on its spec frequency	**Provide the crystal with a specified parallel capacitance** (the load cap). Crystals are spec'd at a particular load capacitance; missing it shifts the frequency slightly.	E7H12
Highly accurate oscillator for microwave work	GPS reference, rubidium reference, or temperature-controlled high-Q dielectric resonator – **all of the above**.	E7H13

Mental Cheat Sheet

- Three classic oscillators: Colpitts, Hartley, Pierce.
- Colpitts → capacitive divider feedback.
- Pierce → quartz crystal feedback.
- Hartley → tapped-coil feedback (third member of the correct E7H01 trio).
- PLL = phase detector + LPF + VCO + reference. Servo loop.
- PLL uses: frequency synthesis and FM demodulation.
- DDS pipeline: phase accumulator → lookup table → DAC → anti-alias filter.
- DDS lookup table holds **amplitude values**.
- DDS impurities: **spurious signals at discrete frequencies**.
- Thermal drift fix in crystal oscillators: NP0 caps.
- Microphonic = vibration-induced frequency change; fix by mechanical isolation.
- Crystal load: provide the **specified parallel capacitance**.
- Highly accurate microwave oscillator: GPS / rubidium / dielectric resonator (all of the above).

Common Traps

- "Three common oscillator circuits" answer is **Colpitts, Hartley, and Pierce**. The test offers fake names (Taft, Fenner, Beane) to throw you off. Memorize the real three.
- A PLL needs all four blocks – phase detector, LPF, VCO, reference. The trap answer is "ratio detector, reactance modulator, VCO" (wrong block names).
- Microphonic = vibration, not microphone preamp or RF pickup.
- DDS lookup table holds **amplitude** values, not phase relationships.
- DDS spurs are at **discrete frequencies**, not broadband.
- The crystal-frequency fix is **parallel capacitance**, not parallel inductance or bias voltage.

Takeaway: Memorize the three classic oscillators and their feedback paths. A PLL is a four-block servo (phase detector + LPF + VCO + reference) used for synthesis and FM demod. DDS turns stored amplitude samples into RF via a DAC. Drift and microphonics are fixed by NP0 caps and mechanical isolation; crystals need their spec load cap to hit the marked frequency.

Bite 8 Question Bank (E7H)

E7H01. What are three common oscillator circuits?
A. Taft, Pierce, and negative feedback.
B. Pierce, Fenner, and Beane.
C. Taft, Hartley, and Pierce.
D. Colpitts, Hartley, and Pierce.

E7H02. What is a microphonic?
A. An IC used for amplifying microphone signals.
B. Distortion caused by RF pickup on the microphone cable.
C. Changes in oscillator frequency caused by mechanical vibration.
D. Excess loading of the microphone by an oscillator.

E7H03. What is a phase-locked loop?
A. An electronic servo loop consisting of a ratio detector, reactance modulator, and voltage-controlled oscillator.
B. An electronic circuit also known as a monostable multivibrator.
C. An electronic servo loop consisting of a phase detector, a low-pass filter, a voltage-controlled oscillator, and a stable reference oscillator.
D. An electronic circuit consisting of a precision push-pull amplifier with a differential phase input.

E7H04. How is positive feedback supplied in a Colpitts oscillator?
A. Through a tapped coil.
B. Through link coupling.
C. Through a capacitive divider.
D. Through a neutralizing capacitor.

E7H05. How is positive feedback supplied in a Pierce oscillator?
A. Through a tapped coil.
B. Through link coupling.
C. Through a neutralizing capacitor.
D. Through a quartz crystal.

E7H06. Which of these functions can be performed by a phase-locked loop?
A. Wide-band AF and RF power amplification.
B. Frequency synthesis and FM demodulation.
C. Photovoltaic conversion and optical coupling.
D. Comparison of two digital input signals and digital pulse counting.

E7H07. How can an oscillator's microphonic responses be reduced?
A. Use NP0 capacitors.
B. Reduce noise on the oscillator's power supply.
C. Increase the gain.
D. Mechanically isolate the oscillator circuitry from its enclosure.

E7H08. Which of the following components can be used to reduce thermal drift in crystal oscillators?
A. NP0 capacitors.
B. Toroidal inductors.
C. Wirewound resistors.
D. Non-inductive resistors.

E7H09. What type of frequency synthesizer circuit uses a phase accumulator, lookup table, digital-to-analog converter, and a low-pass anti-alias filter?
A. A direct digital synthesizer.
B. A hybrid synthesizer.
C. A phase-locked loop synthesizer.
D. A direct conversion synthesizer.

E7H10. What information is contained in the lookup table of a direct digital synthesizer (DDS)?
A. The phase relationship between a reference oscillator and the output waveform.
B. Amplitude values that represent the desired waveform.
C. The phase relationship between a voltage-controlled oscillator and the output waveform.
D. Frequently used receiver and transmitter frequencies.

E7H11. What are the major spectral impurity components of direct digital synthesizers?
A. Broadband noise.
B. Digital conversion noise.
C. Spurious signals at discrete frequencies.
D. Harmonics of the local oscillator.

E7H12. Which of the following ensures that a crystal oscillator operates on the frequency specified by the crystal manufacturer?
A. Provide the crystal with a specified parallel inductance.
B. Provide the crystal with a specified parallel capacitance.
C. Bias the crystal at a specified voltage.
D. Bias the crystal at a specified current.

E7H13. Which of the following is a technique for providing highly accurate and stable oscillators needed for microwave transmission and reception?
A. Use a GPS signal reference.
B. Use a rubidium stabilized reference oscillator.
C. Use a temperature-controlled high Q dielectric resonator.
D. All these choices are correct.

Answer Key

E7H01: D	E7H02: C	E7H03: C	E7H04: C
E7H05: D	E7H06: B	E7H07: D	E7H08: A
E7H09: A	E7H10: B	E7H11: C	E7H12: B
E7H13: D			